

Reducing Customer Churn Through Predictive Analytics

A data-driven retention program to protect at-risk subscriber revenue

Prepared by: Nyasha Chinyanga - Data Analytics
Prepared for: Executive Leadership & Customer Retention Teams
Based on analysis of 100,000 customer accounts, usage records, and churn outcomes

49.6% of customers churned	\$2.88M monthly revenue at risk	91% of churners caught by the model	\$34.6M annualized revenue exposure
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Executive Summary

Nearly half of the customer base analyzed in this study churned within the observed period, representing \$2.88 million in monthly recurring revenue, roughly \$34.6 million annualized at risk. This proposal recommends deploying a predictive churn model, built and validated on this customer data, to identify at-risk customers before they leave and route them into a targeted retention program.

The model identifies the customers most likely to churn with 91% recall, using only data the business already collects: usage patterns, billing history, equipment age and account tenure. No new data infrastructure is required to act on it. Directing retention offers toward the customers this model flags, rather than running blanket, unfocused campaigns. The model is projected to protect between \$144,000 and \$577,000 in monthly revenue, depending on the retention team's campaign success rate.

The Problem

Customer churn is not a minor leakage problem here, it is close to a coin flip, of 100,000 customer accounts analyzed, 49.6% churned. Three findings make this more urgent than the headline rate alone suggests:

- Churn is broad-based, not a fringe issue:** across the customer base, customers of every spending level churn at a similar rate (48 - 53%) , this is not a problem concentrated in low-value, easy-to-write-off accounts.
- Revenue at risk is concentrated at the top:** the top revenue quartile alone accounts for \$1.39 million of the \$2.88 million at risk. Losing a high-value customer costs far more than losing a low-value one, even though both are equally likely to leave.
- Churn risk climbs steadily with equipment age:** customers with handsets older than three years churn at 60%, compared to 45% for customers on devices under a year old, a visible, actionable early-warning signal sitting in data the business already has.

Without a way to identify which customers are at risk before they leave, retention spend is necessarily reactive or untargeted, either responding after a customer has already decided to leave, or spread thinly across a customer base where only half will ever actually churn.

Our Approach

We combined two internal data sources (customer usage/billing records and account-level demographic data) into a single customer view spanning 100,000 accounts and over 120 attributes. From there:

- Feature engineering:** Beyond raw usage totals, we engineered behavioral indicators, cost per minute, recent-vs-historical usage trend, service call-failure rate and customer-care friction that describe how a customer's relationship with the service is changing not just how much they use it.

- **Feature selection:** Using two independent statistical methods, we narrowed over 145 candidate attributes down to the 25 most predictive, non-redundant features keeping the model focused and its findings easy to explain.
- **Model development:** We trained and tested a Catboost model.
- **Validation:** Every finding in this proposal is drawn directly from validated results on held-out data not assumptions.

What Drives Churn: Key Findings

The model's own feature-importance ranking, cross-checked against two independent statistical measures, points to five consistent drivers:

Driver	What it tells us
Equipment age & tenure	The single strongest signal. Churn rises steadily the longer a customer keeps their current handset an upgrade or loyalty touchpoint at the right moment is a natural intervention point.
Usage trend ratios	Customers whose recent usage is dropping relative to their own historical average are disengaging before they formally churn a leading indicator, not a lagging one.
Cost / pricing efficiency	What a customer pays per minute of use relative to their plan matters more than their raw bill size, a signal for plan-fit conversations rather than blanket discounting.
Service call-failure rate	Dropped and blocked calls are a direct service-quality signal and a low-cost trigger for proactive outreach.
Revenue & lifetime-value trend	Declining revenue and lifetime-value trajectory rounds out the picture of a customer whose relationship with the service is weakening.

The Model: Performance & Trade-offs

The recommended model is a CatBoost classifier, selected after head-to-head testing against Random Forest and XGBoost under identical 5-fold cross-validation:

Model	Mean AUC	Mean Recall
Random Forest	0.676	63.7%
XGBoost	0.669	64.6%
CatBoost (recommended)	0.681	91.4%

CatBoost was tuned and its decision threshold optimized specifically to maximize recall its ability to correctly flag customers who are actually going to churn rather than raw accuracy. This is a deliberate business trade-off, not a modeling oversight: in a retention context, the cost of missing a real churner (a lost customer) is far higher than the cost of a false alarm (one unnecessary, low-cost retention touchpoint to a customer who was never going to leave). The model currently catches roughly 91% of customers who go on to churn.

Business Impact: Revenue Opportunity

Applying the model to flag at-risk customers turns a monthly revenue-at-risk figure of \$2.88 million into an actionable, prioritized retention list. The revenue protected depends on how successful the retention team's outreach is at actually keeping a flagged customer as illustrated below across a realistic range:

Retention campaign success rate	Monthly revenue protected	Annualized
5%	\$144,000	\$1.73M
10%	\$288,000	\$3.46M
15%	\$432,000	\$5.19M
20%	\$576,000	\$6.92M

Even a conservative 10% retention success rate on flagged customers, achievable with a well-targeted campaign represents roughly \$3.5 million in protected annual revenue at a fraction of the cost of an untargeted, blanket retention effort.

Recommended Implementation Plan

- **Phase 1 Deploy (Weeks 1–4):** Score the full active customer base monthly using the validated model, output a ranked, prioritized risk list.
- **Phase 2 Integrate (Weeks 3–6):** Feed the top-risk, high-value segment into the retention team's existing CRM workflow, paired with the specific driver(s) behind each customer's risk score (e.g. aging equipment, declining usage, service friction).
- **Phase 3 Pilot (Weeks 6–12):** Launch driver-matched interventions, device upgrade offers for equipment-age risk, plan review for cost-efficiency risk, proactive service follow-up for call-failure risk on a defined pilot segment.
- **Phase 4 Measure & Scale:** Compare actual churn among the pilot's flagged-and-contacted customers against a held-out control group, confirm the real-world retention success rate and refine the revenue-impact estimate before scaling company-wide.

Conclusion

This analysis shows that churn, while widespread, is not unpredictable, the signals that precede it (aging equipment, softening usage, service friction and pricing mismatch) are already sitting in existing data. The recommended model surfaces those signals early enough to act on them, with 91% of eventual churners identified in advance. We recommend approving a pilot along the plan above, sized to validate the revenue-protection estimates in this proposal against real retention outcomes before a full rollout.